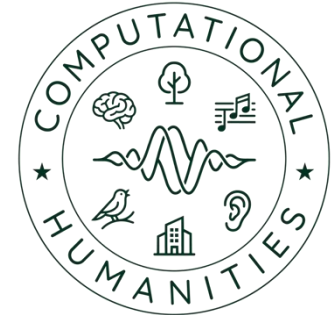


University of Bamberg



Master Project - Machine Listening (CH-Proj-M) 00 - Introduction

Prof. Dr.-Ing. Jakob Abeßer
Professur für Computational Humanities
Fakultät für Wirtschaftsinformatik und Angewandte Informatik
Otto-Friedrich-Universität Bamberg, 96047 Bamberg
E-Mail: jakob.abesser@uni-bamberg.de
Web: <https://www.uni-bamberg.de/ch/>

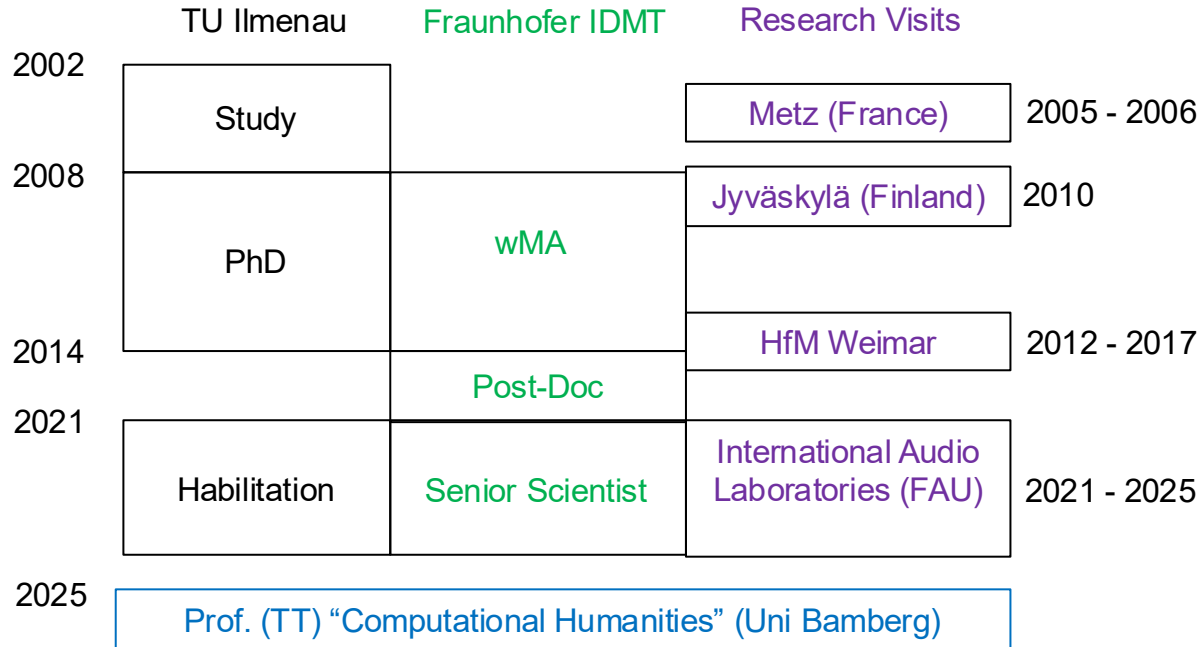


Bio

Prof. Dr. Jakob Abeßer



B0



Introduction

Computational Humanities (Uni Bamberg)



- Website
 - <https://www.uni-bamberg.de/ch/>

The screenshot shows the website of the Chair for Computational Humanities. The header is dark blue with the University of Bamberg logo and name. Below the header, there's a light blue bar with the faculty name and the chair's name. A navigation menu on the left lists: Start, Team, Forschung, Studium, Publikationen, and News. The main content area has the title 'Professur für Computational Humanities' and a paragraph describing the chair's focus on the intersection of informatics and humanities, specifically in sound research. Below this, it mentions research focus areas like digital musicology and soundscape research. A 'News' section on the right highlights a project 'ERBA-SOUND' and a participation in a conference 'Akustische Umwelt mit KI gestalten'.

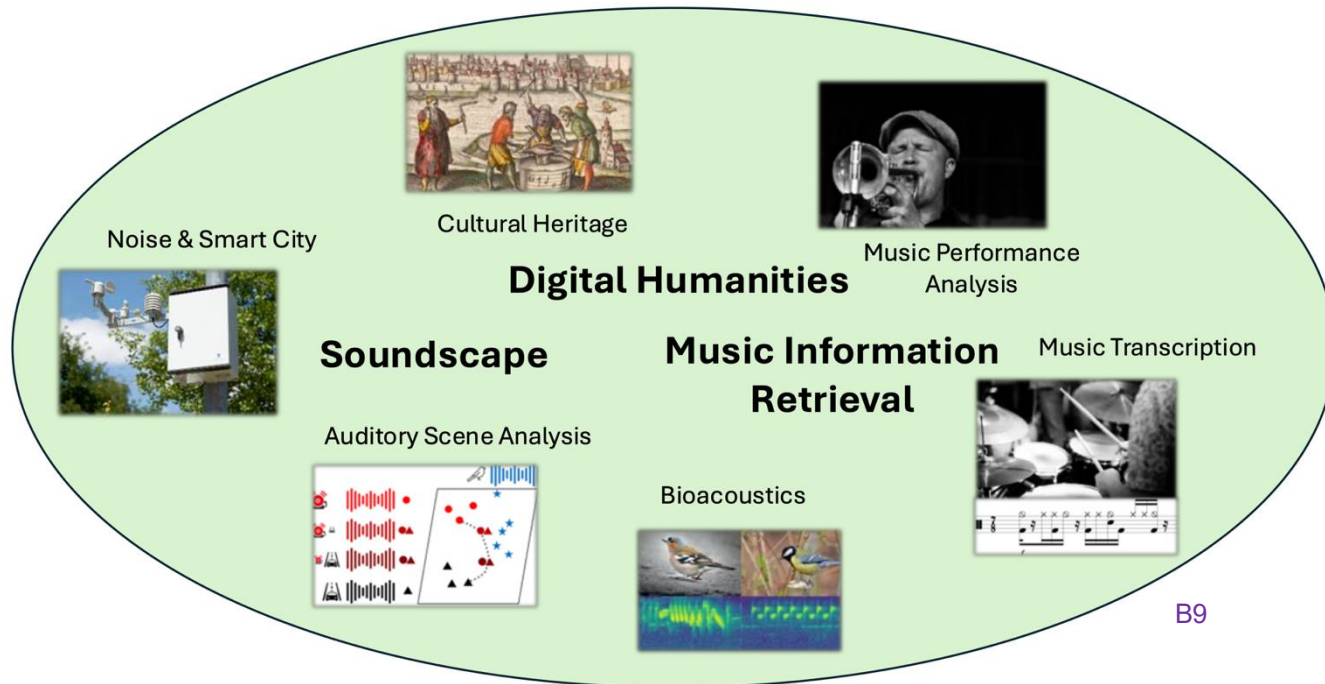


Introduction

Computational Humanities (Uni Bamberg)



- Research



Introduction

Computational Humanities (Uni Bamberg)



- Teaching (SS 2026)
 - [CH-EAM-B \(Foundations of Audio and Music Processing\)](#)
 - [CH-CMP-B \(Computer-Assisted Music Production: Algorithms, Sound Design, and Composition\)](#)

Lecture

Concept



- 6 ECTS
- Every week
 - 2 blocks (10:15-11:45, 12:15-13:45, WE5/01.003)
 - 15–30-minute pitch presentation (various topics)
 - Students work on the selected project within defined project phases
 - Supervision available on-site
- Project
 - 1-2 students per team

Lecture

Concept



- VC Course
 - <https://vc.uni-bamberg.de/course/view.php?id=81301>
 - Regular announcements
 - Slides
 - Optional reading material

Lecture Structure



Week	Day	Project Phase	Topics
1	13.04.		Overview, master project organization, open project topics
2	20.04.	1. Literature research	Literature research strategies, reference management
3	27.04.		Basics: academic writing, required structure of the term paper
4	04.05. (P1)	2. Data acquisition (Dataset research, data recording)	Role of data and annotations in machine listening research, data protection, licences (CC etc.)
5	11.5.		Basics: audio editing and annotation (Audacity, Sonic Visualizer)

Lecture Structure



6	18.5. (P2)	3. Baseline system (Audio analysis + machine learning method)	Basics: time-frequency representations & audio features
7	25.05. (Holiday)		-
8	01.06.		Basics: machine learning & evaluation strategies & metrics
9	08.06.	4. Advanced System (Extension using deep learning method)	Basics: deep learning
10	15.06.		tbd.
11	22.06.		tbd.
12	29.06.	Possible field trip (Fraunhofer IDMT)	tbd.
13	06.07.	Completion of term paper & presentation	Deadline paper submission (6.7., EOB)
14	13.07. (P3)	5. Project presentation	

Lecture

Requirements & Grading



- Presentations (First block, starting Monday, 10:15)
 - **P1 (04.05.):** 5+5 min (Q/A): topic + literature review
 - **P2 (18.05.):** 5+5 min (Q/A): data set / data acquisition strategy
 - **P3 (13.07.):** 10+5 min (Q/A): final project presentation
- Term paper (**deadline: 06.07. EOB**)
 - 4-6 pages, structure & requirements will follow later
- **Project grading**
 - **Mandatory: 3 presentations + paper**
 - **Project grade: P1: 15% + P2: 15% + P3: 35% + Paper: 35%**

Lecture

Concept



- Field trip (optional): **29.06.2026**
 - [Fraunhofer Institute for Digital Media Technologies \(IDMT\)](#)
 - Ilmenau (Thüringen): [Location](#)
 - Travel by train:
 - Bamberg > Erfurt > Ilmenau



Lecture

Concept

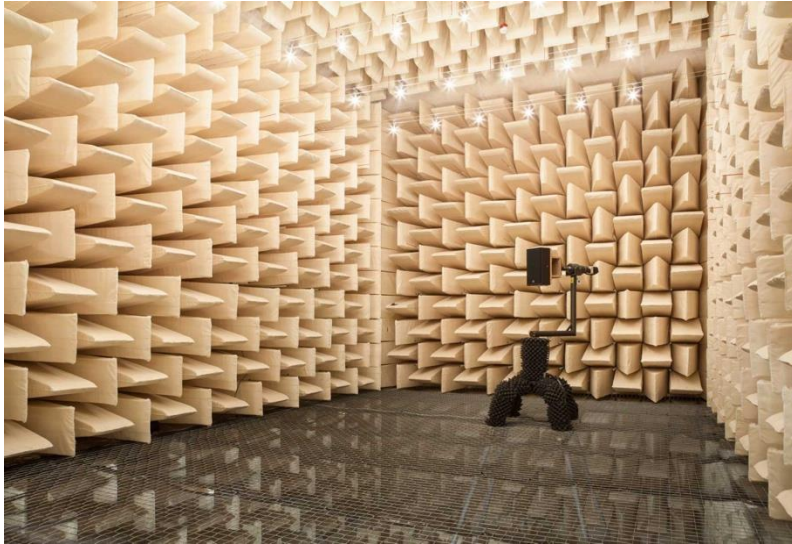


- Insight into the work of a leading applied research institute in Audio + AI
- Preliminary presentation of project results to subject matter experts
- Meet various research groups (Semantic Media Technologies, Industrial Media Applications, Smart Audio Solutions, and Media Distribution and Security groups)

Lecture

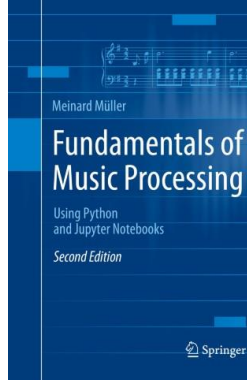
Concept

- Anechoic chamber, Spatial audio demos

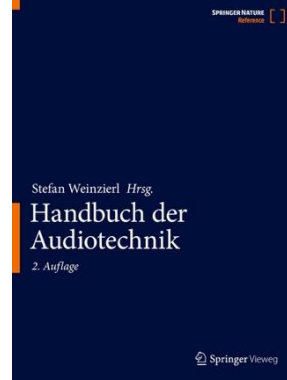


Resources

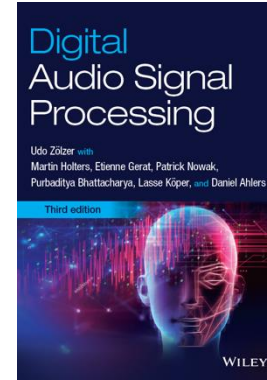
Books



B10



B11



B12

- Müller, M.: *Fundamentals of Music Processing – Using Python and Jupyter Notebooks*, Springer, 2021.
- Weinzierl, S. (Ed.). (2025). *Handbuch der Audiotechnik* (2nd ed.). Springer.
- Zölzer, U. (2022). *Digital audio signal processing* (3rd ed.). John Wiley & Sons Ltd.

Resources

Websites



- Python
 - Preparation Course Python <https://audiolabs-erlangen.de/PCP>
- Digital Signal Processing
 - Digital Signals Theory (online book)
<https://brianmcfee.net/dstbook-site/content/intro.html>

Resources

Websites



- Music Information Retrieval
 - <https://www.audiolabs-erlangen.de/FMP> (iPython notebooks)
 - <https://musicinformationretrieval.com> (iPython notebooks)

Resources

Python Packages

- Numerical computing / Statistics / Visualization
 - [numpy](#), [scipy](#), [pandas](#), [matplotlib](#), [seaborn](#)
- Machine Learning / Deep Learning
 - [scikit-learn](#), [tensorflow](#), [pytorch](#)
- Audio & Music Processing
 - [librosa](#), [soundfile](#), [pysox](#), [audiomentations](#)
 - [madmon](#), [libfmp](#), [synctoolbox](#), [libtsm](#)

Image Sources



B0: © Tim Kipphan/Universität Bamberg

B1: <https://unsplash.com/de/fotos/eine-grosse-gruppe-weisser-und-schwarzer-buchstaben-aNVx2FpKAF8>

B2: <https://unsplash.com/de/fotos/strasse-mit-verschiedenen-autos-m2TKgpU3cvc>

B3: <https://unsplash.com/de/fotos/orchester-spielen-ihr-stuck-7CcPLtywRso>

B4: <https://unsplash.com/de/fotos/junge-sitzt-auf-weissem-tuch-umgeben-von-spielzeug-p0hDztR46cw>

B5: https://unsplash.com/de/fotos/menschen-die-tagsuber-bechertrommeln-spielen-OJ4m4_ylgjY

B6: <https://unsplash.com/de/fotos/personen-die-tagsuber-auf-der-strasse-vorbeifahren-UligFW4P7L8>

B7: <https://unsplash.com/de/fotos/bauarbeiter-auf-der-strasse-WJg2bynUWOk>

B8: https://unsplash.com/de/fotos/tierfotografie-flamingoschwarm-ub1sSvJ_Tbs

Image Sources



B9:

- <https://unsplash.com/de/fotos/person-die-schlagzeug-spielt-nxp5B03CXkE>
- https://de.wikibooks.org/wiki/Pythagoras_in_der_Schmiede
- <https://unsplash.com/de/fotos/graustufenfoto-eines-musikers--GsPtETXbfo>
- Abeßer, et al., 2019, p. 2, fig. 2
- Abeßer, et al., 2023, p. 2659, fig. 1
- Abeßer, et al., 2025, p. 13, fig. 5 (modified)

B10: <https://media.springernature.com/w316/springer-static/cover-hires/book/978-3-030-69808-9?as=webp>

B11: https://www.hsu-hh.de/ant/wp-content/uploads/sites/699/2022/10/DASP_3rd_cover_preferenc-2021-11-01.png

B12: https://media2.hugendubel.de/shop/coverscans/376/37612518_9783662603680_xl.jpg

Sound Sources



A1: Medley (<https://freesound.org/people/FunWithSound/sounds/416645/>,
<https://freesound.org/people/GR3AVE5Y/sounds/149109/>,
<https://freesound.org/people/blaukreuz/sounds/212194/>,
<https://freesound.org/people/RTB45/sounds/157642/>,
<https://freesound.org/people/trouby/sounds/362387/>)

Thank you for your attention!

Any questions?



Prof. Dr. Jakob Abeßer

jakob.abesser@uni-bamberg.de

<https://www.uni-bamberg.de/ch/>